**PVP23**

Code No:**23ES1305**

# PVP SIDDHARTHA INSTITUTE OF TECHNOLOGY

**(Autonomous)**

**ARTIFICIAL INTELLIGENCE**

**(CSE (AI&ML)**

**Model Paper**

**Duration: 3 Hours** **Max. Marks: 70**

Note:

1. This question paper contains two parts A and B.
2. Part-A contains 10 short answer questions. Each question carries 2 marks.
3. Part- B contains 5 essay questions with an internal choice from each unit. Each question carries 10 marks.
4. All parts of Question paper must be answered in one place.

BL-Bloom's Level CO-Course Outcome

**PART-A** 10 x 2 = 20 Marks

| | | Blooms Level | CO | Max. Marks |
|---|---|---|---|---|
| 1 | Explain why chess game is semi dynamic | L2 | CO1 | 2 |
| 2 | Differentiate Rationality and Omni Science | L2 | CO1 | 2 |
| 3 | Explain Zero-sum game with an example | L2 | CO1 | 2 |
| 4 | Explain about Search algorithms | L2 | CO1 | 2 |
| 5 | Differentiate Propositional and First-Order Inference | L2 | CO1 | 2 |
| 6 | Explain Inference rules for quantifiers | L2 | CO1 | 2 |
| 7 | Define optimistic description | L2 | CO1 | 2 |
| 8 | How to represent set of ground actions | L2 | CO1 | 2 |
| 9 | Discuss Theory of Learning to restaurant problem | L2 | CO1 | 2 |
| 10 | Describe Q-learning | L2 | CO1 | 2 |

**PART –B** 5 X 10 =50

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|---|---|---|---|---|
| | **UNIT-I** | | | |
| 1 | Describe the foundations of artificial intelligence | L2 | CO1 | 10 |
| | **OR** | | | |
| 2 | Summarize the history of AI. | L2 | CO1 | 10 |
| | **UNIT-II** | | | |
| 3 | Apply Hill Climbing Algorithm to solve 8-Queeens Problem | L3 | CO2 | 10 |
| | **OR** | | | |
| 4 | Analyze BFS and DFS to evaluate Completeness, Space complexity ,time complexity and path cost | L4 | CO4 | 10 |
| | **UNIT-III** | | | |
| 5 | Illustrate the syntax and semantics of First Order Logic. | L3 | CO2 | 10 |
| | **OR** | | | |
| 6 | Analyze forward chaining algorithm with example | L4 | CO4 | 10 |
| | **UNIT-IV** | | | |
| 7 | Illustrate Heuristics for planning. | L3 | CO3 | 10 |
| | **OR** | | | |
| 8 | Apply Classical Planning with Air cargo Transport Example | L3 | CO3 | 10 |
| | **UNIT-V** | | | |
| 9 | Explain the following terms:<br>a. Univariate linear regression<br>b. Multivariate linear regression | L2 | CO1 | 10 |
| | **OR** | | | |
| 10 | Apply Passive Reinforcement Learning with an example | L3 | C03 | 10 |